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A Data-Driven Visualization and Prediction Study of Conflict Impacts on Education and Food Security in Gaza

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Abstract

This graduation project presents a data-driven analysis of the impact of armed conflict on food security and education in Gaza, utilizing incident-level humanitarian data and media-based information. The study integrates two primary datasets obtained from the Humanitarian Data Exchange (HDX), documenting conflict incidents affecting food systems (2023–2025) and education (2020–2024). The main objective of the project is to analyze spatial, temporal, and sector-specific patterns of conflict impacts and to support evidence-based insights for decision-makers through advanced visualization and analytical techniques.

The project implementation followed a structured Business Intelligence workflow. Raw datasets were extracted in Excel format, cleaned, transformed, and analyzed using Power BI to develop interactive dashboards focused on food security and education impacts. Geospatial maps, line charts, bar charts, and KPI cards were employed to highlight trends, hotspots, and severity levels. In parallel, an advanced analytics component was developed using Python, applying a hybrid Natural Language Processing (NLP) approach that combines semantic similarity modeling and keyword-based matching to identify food security–related news articles from Al Jazeera via the GDELT API. Multilingual sentence embeddings were used to support both Arabic and English content, enabling scalable and interpretable classification of relevant media coverage.

The results reveal clear temporal fluctuations and geographic concentrations of conflict-related incidents, with aid systems and school-level education emerging as the most affected components. The NLP model successfully captured both explicit and implicit references to food security, enhancing coverage beyond traditional keyword filtering. Overall, the project demonstrates how integrating humanitarian data, media analytics, interactive visualization, and AI-based methods can provide a comprehensive and forward-looking assessment of conflict impacts on essential civilian systems, supporting data-driven humanitarian and policy decision-making.

Acknowledgment

I would like to express my sincere gratitude to everyone who contributed to the completion of this graduation project.

First and foremost, I would like to extend my deepest appreciation and special thanks to **Dr. Husam Barham** for his exceptional guidance, continuous support, and genuine dedication throughout this project. His commitment, encouragement, and willingness to stand by us at every stage made a meaningful difference in our academic journey. His valuable insights and constructive feedback were instrumental in shaping this work and ensuring its successful completion.

I am also deeply grateful to the faculty members of the Administrative and Financial Sciences Department for their academic guidance, support, and insightful feedback, which greatly enriched the quality of this project.

Special thanks go to my colleagues for their cooperation, idea-sharing, and teamwork throughout this course. Their collaboration and mutual support added significant value to this experience.

I would like to express my heartfelt gratitude to my family for their endless love, patience, and unwavering belief in me. Your support has been my greatest source of strength and motivation throughout this journey.

Finally, I extend my sincere thanks to my friends and to everyone who supported me, directly or indirectly. Your encouragement and presence have been invaluable, and I am truly grateful.

Business Intelligence Project description and objectives

This graduation project is based on the analysis of **two structured datasets** obtained from **open humanitarian data sources**, both documenting the impact of armed conflict on essential civilian systems.

The first dataset, titled **“2023–2025 PSE Gaza Conflict Incidents Affecting Food Systems,”** includes **more than 1,000 incident-level records** related to the Gaza conflict.
It provides detailed information on the **date, geographic location, weapon type, reported perpetrator, and the impact on food systems**, including disruptions to food supply chains, humanitarian aid systems, and people involved in food-related activities.
This dataset enables in-depth analysis of how conflict dynamics affect food security and access to food.

The second dataset focuses on **incidents affecting education** between **2020 and 2024**, covering attacks on **schools, universities, students, and educators** across multiple countries.
This dataset includes **282 recorded incidents**, capturing information on **incident timing, geographic location, type of attack, responsible actors, and the consequences for educational facilities and individuals**, including killings, injuries, arrests, and kidnappings.
The dataset supports risk analysis, reporting, and evidence-based assessments related to the protection of education in conflict-affected and fragile contexts.

Project Description (Short Summary):

Together, these two datasets form the analytical foundation of the project. They are used for data cleaning, exploratory analysis, geospatial visualization, and dashboard development, enabling a comprehensive assessment of how armed conflict affects food security and access to education. The analysis focuses on identifying historical patterns, spatial concentrations, and temporal trends, supporting evidence-based interpretation and data-driven decision-making.

Tools and Technologies:

- Microsoft Excel
- Power BI
- Python

Data Research and Acquiring Effort

- The datasets used in this project were obtained from the **Humanitarian Data Exchange (HDX)** and provided by **Insecurity Insight**.
  They include **1,005 incident records related to food systems affected by conflict in Gaza between 2023 and 2025**, as well as **282 incident records related to education affected by conflict**, including schools, universities, students, and educators, recorded between **2020 and 2024**.
  Both datasets were selected due to their **high reliability, open licensing, standardized incident-level structure, and strong relevance to conflict and humanitarian impact analysis**, making them suitable for academic research, data analysis, and evidence-based assessment.

  https://data.humdata.org

Links to raw data

- **Incident-level data on the impact of the Gaza conflict on food systems**

2023-2025-pse-gaza-conflict-incidents-affecting-food-systems-incident-data-incident-data.xlsx

- **Incident data documenting the impact of conflict on the education sector in Palestine**

2020-2024-pse-education-in-danger-incident-data.xlsx

- **An interactive Power BI dashboard analyzing the impact of the war on food and education in Gaza, with integrated news data to provide contextual insights**.

Gaza_Food_and_Education_Impact_Dashboard.pbix

Data Description and understandings

This project is based on the analysis of two incident-level datasets documenting the impact of armed conflict on **food systems** and **education**. In both datasets, each row represents a single recorded incident, while each column represents a specific attribute describing the time, geographic location, responsible actor, method used, and sectoral impact of the event.

The first dataset covers incidents that occurred between **2023 and 2025** and focuses on events affecting food systems in Gaza. Each incident is uniquely identified using an **Event ID**, while the **Date** column specifies the exact day of occurrence. The **Country** and **Country ISO (PSE)** fields confirm that all recorded incidents took place within the Occupied Palestinian Territory (State of Palestine). Geographic variables such as **Admin 1**, **Latitude**, and **Longitude** enable precise spatial visualization, while the **Geo Precision** field indicates the level of accuracy of the recorded location, which is essential for reliable geospatial analysis.

Responsibility and execution details are captured through the **Reported Perpetrator** and **Reported Perpetrator Name** fields, which identify the actors involved, including state military forces. The **Weapon Carried/Used** variable records the type of weapon or method employed, such as firearms, drones, or explosives. A key analytical variable in this dataset is **Food System Impact**, which specifies the affected component of the food system, including food supply, aid systems, or people carrying out food-related activities. In cases where multiple impacts were associated with a single incident, these were separated during the data-cleaning process to ensure analytical accuracy. Additional contextual information is provided through the **All Food Security Categories** field, while the **Date Event Entered** and **Date Event Modified** fields support data transparency, traceability, and reliability.

The second dataset documents **education-related incidents** that occurred between **2020 and 2024**, capturing attacks targeting schools, universities, students, and educators in conflict-affected contexts. Each incident is uniquely identified using the **SiND Event ID** and includes standardized geographic variables such as **Country**, **Country ISO**, **Admin 1**, **Latitude**, **Longitude**, and **Geo Precision**, enabling both spatial and comparative analysis.

This dataset provides detailed information on the nature of attacks on educational infrastructure through variables such as **Type of Education Facility**, **Attacks on**

Schools, **Attacks on Universities**, **Military Occupation of Education Facility**, **Forced Entry**, **Arson Attacks**, and **Damage or Destruction of Educational Facilities**. It also captures the human impact of conflict by documenting harm to educators and students, including **killings, injuries, kidnappings, and arrests**, along with follow-up outcome fields where information is available. Furthermore, incidents involving **sexual violence affecting school-age children** are explicitly flagged, highlighting severe violations within educational settings. As with the food systems dataset, the **Date Event Entered** and **Date Event Modified** fields ensure traceability and support sound data governance practices.

Overall, both datasets provide comprehensive and structured records of conflict-related incidents, capturing their timing, geographic location, responsible actors, methods used, and sector-specific impacts. Their combined use enables **Business Intelligence analysis, geospatial visualization, trend analysis, and predictive modeling**, supporting a holistic assessment of how armed conflict affects food security and access to education.
It is important to note that the datasets capture **documented incidents specifically affecting food systems and education**, and therefore **do not represent the total volume of conflict events or casualties**, but rather focus on sector-specific impacts relevant to the scope of this study

Data Primary Cleaning and Transformation

The data preparation phase represents a critical foundation for ensuring the reliability and validity of the analytical outcomes in this project. The raw datasets were imported into Power BI using the Get Data functionality, after which all preprocessing tasks were conducted within Power Query Editor following a structured and sequential approach.

Initially, an exploratory data review was performed to assess the overall structure and quality of the datasets. This included identifying missing values, duplicated records, inconsistent column formats, and potential data integrity issues. Based on this assessment, all columns were systematically converted to their appropriate data types, such as Whole Number, Decimal Number, Text, and Date. This step was essential to guarantee accurate aggregations, calculations, and visual representations in subsequent analytical stages.

Following data type standardization, comprehensive data cleaning procedures were applied. Rows containing null or incomplete values that could negatively impact the analysis were removed. In addition, duplicated columns and redundant records were eliminated to enhance dataset clarity and analytical precision. General cleaning techniques were also employed, including text normalization, trimming unnecessary spaces, and correcting formatting inconsistencies, ensuring a consistent and analysis-ready dataset.

To further support analytical interpretation and insight generation, additional calculated and custom columns were created. These columns were designed to enrich the dataset by introducing derived attributes, classifications, or indicators that improve contextual understanding and enable more meaningful segmentation and comparison. The creation of these columns played a key role in simplifying report development and aligning the dataset with the project's analytical objectives.

Once the data was cleaned and enriched, multiple datasets were consolidated using the Append Queries method. The decision to apply appending rather than Merge operations or table relationships was driven by the absence of a reliable primary key and the high level of duplicated values across datasets. Implementing merge logic or relationships under these conditions could have resulted in many-to-many relationships, record duplication, or inaccurate aggregations. Therefore, appending the datasets into a unified table structure was the most appropriate approach to preserve data integrity and ensure consistent analysis.

Finally, the transformed dataset was validated by reviewing row counts, checking for transformation errors, and confirming data consistency. The finalized dataset was then loaded into the Power BI data model, providing a robust and reliable foundation for building dashboards, visual analytics, and data-driven insights aligned with the objectives of the graduation project

Data Visualization and Insights

Dashboard Overview

This dashboard was designed to visualize and analyze the **impact of war on the Gaza Strip**, with a particular focus on the **food and education sectors**. The data is presented through a combination of **charts, geospatial maps, and interactive filtering tools**, supporting both **high-level monitoring** and detailed data exploration.

The dashboard enables users to identify key patterns and trends related to incidents across time, location, and impact type, facilitating data-driven analysis and insights. It was designed with a **clear and visually accessible layout**, incorporating **colors inspired by the Palestinian flag** and the **watermelon symbol**, which represents Palestinian identity and resilience, while maintaining an academic and professional presentation

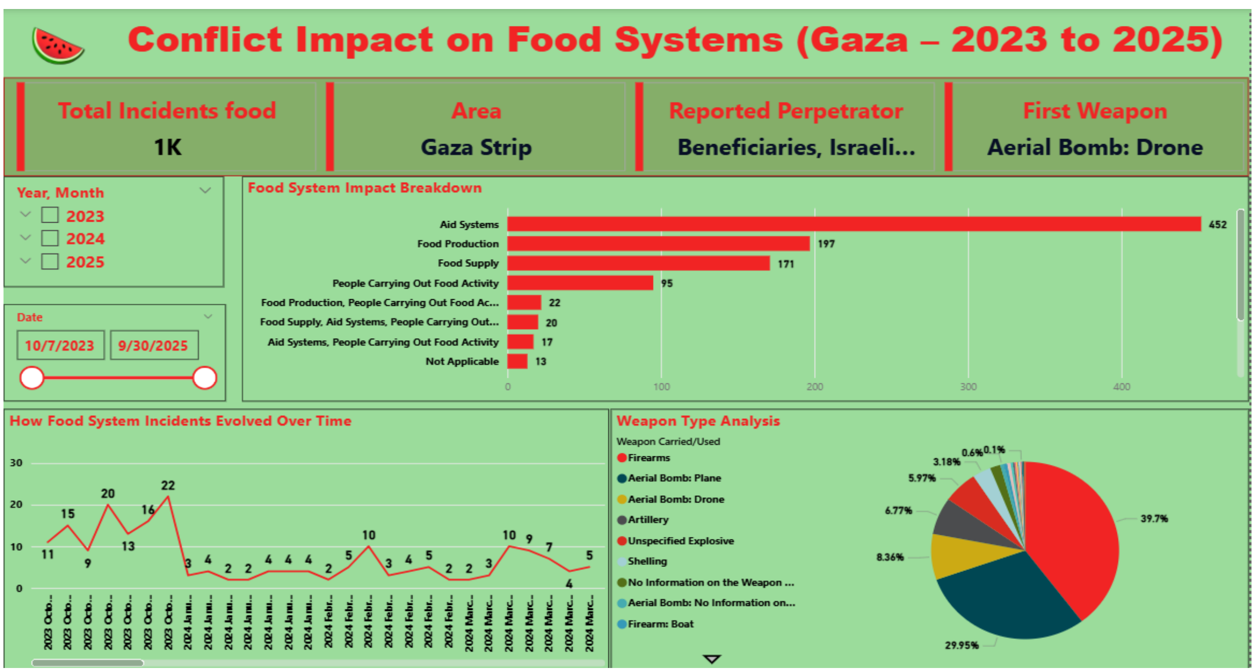

Figure 1: Conflict Impact on Food Systems

Dashboard KPI Cards

The dashboard includes a set of KPI cards designed to provide a quick and high-level summary of key information related to food-related incidents. These cards allow users to immediately understand the overall situation without navigating detailed charts.

Card 1 – Total Incidents (Food): Displays the total number of recorded food-related incidents, providing a clear overview of the scale of impact.

Card 2 – Area: Identifies the affected geographical area. The field was renamed from Admin 1 to Area to improve clarity and ensure better understanding for non-technical users.

Card 3 – Reported Perpetrator: Shows the entity reported or associated with the incident, supporting accountability and contextual analysis.

Card 4 – First Weapon Used: Indicates the initial weapon used in the incident, offering insight into the nature and severity of the event.

Date Slicers (Time Filtering)

Two date slicers were implemented to enhance time-based analysis and user interactivity. The first slicer allows users to filter incidents by Year and Month, enabling high-level temporal exploration and comparison across different periods. The second slicer is a manual date range slicer, which provides more precise control by allowing users to define a specific start and end date.
Together, these slicers improve analytical flexibility, allowing both broad trend analysis and detailed time-specific investigation.

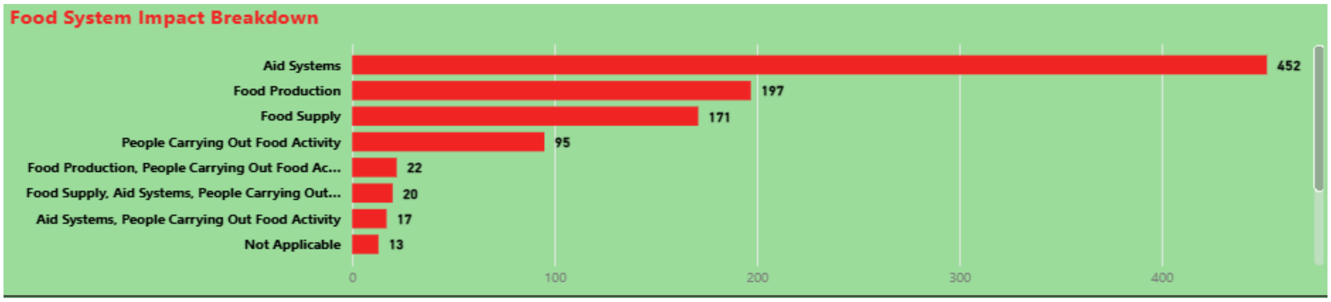

Figure 2: Food System Impact Breakdown 1

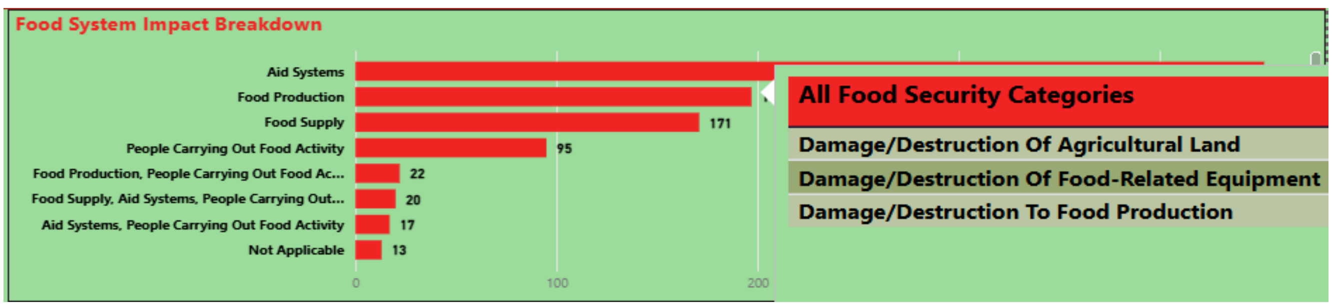

Figure 3: Food System Impact Breakdown 2

Food System Impact Breakdown

A clustered bar chart was used to display the distribution of food system impacts based on the count of recorded incidents. The chart shows that **aid systems** experienced the highest level of impact compared to other food system components. A tooltip was added to provide additional context by displaying the related **Food Security Categories**, such as damage or destruction of agricultural land, food-related equipment, and food production.

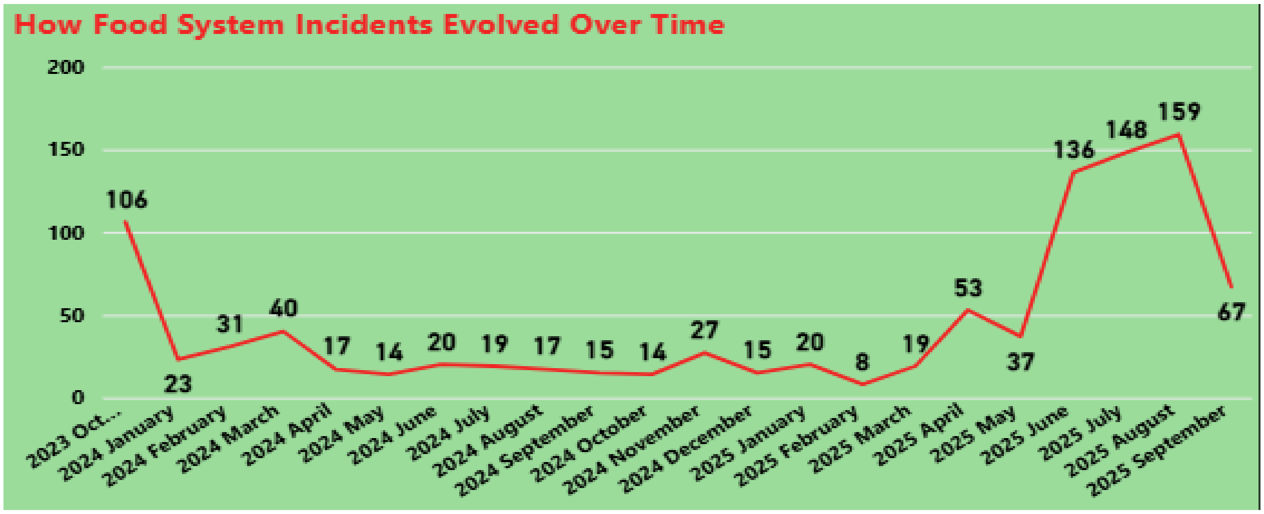

Figure 4: How Food System Incidents Evolved Over Time

How Food System Incidents Evolved Over Time

This line chart shows how food system incidents changed over time using the Date (Year/Month) dimension. It helps identify periods of escalation and decline, supporting time-based monitoring of impact. The chart highlights clear fluctuations, with noticeable peaks that indicate months with significantly higher incident levels, which can guide stakeholders in pinpointing high-risk periods.

To enhance time-based analysis, a new column Day Name was created using DAX to convert the date into weekday names (FORMAT(Date,"dddd")). In addition, a DayOrder column was generated using SWITCH to assign a logical weekday order (Saturday to Friday). This ensures correct sorting and consistent reporting when analyzing incidents by day of the week.

Insight: The visualization enables identifying when damage was most intense over time, supporting planning, response prioritization, and resource allocation during peak periods

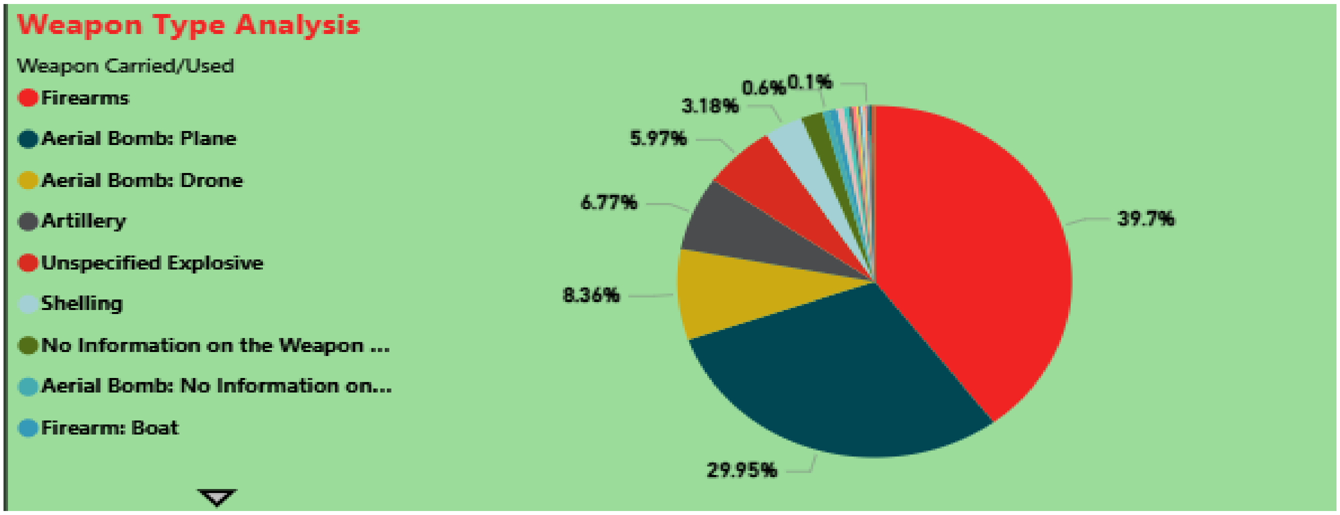

Figure 5: Weapon Type Analysis

Weapon Type Analysis

This pie chart illustrates the distribution of weapon types used during conflict, calculated as a percentage of total events (Event ID). The visualization highlights the relative frequency of each weapon type, allowing for a clear comparison of usage patterns during the war.

The results indicate that certain weapon types account for a higher proportion of recorded events, while a noticeable share is classified as “No Information on the Weapon Used.” This reflects the challenges associated with data collection and verification during conflict situations, where accurate documentation of weapon usage is often difficult.

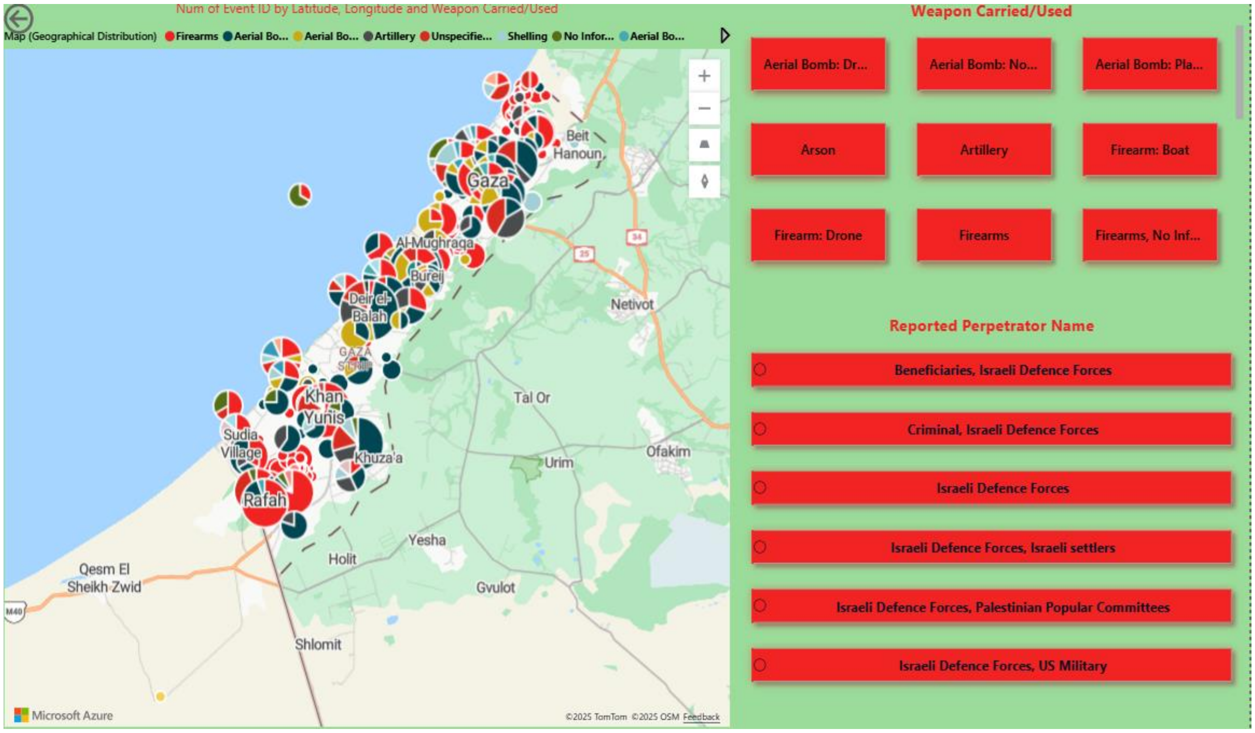

Figure 6: Map of Geographical Distribution of Incidents

Geographical Distribution of Incidents

This map visualization displays the geographical distribution of food system incidents using latitude and longitude, with the count of Event ID representing incident frequency. The map highlights spatial concentration across different locations, allowing users to identify areas with higher levels of impact.

Interactive checkbox slicers were applied for both weapon type and reported perpetrator, enabling users to dynamically filter incidents based on specific criteria. This interactivity supports deeper spatial analysis by linking geographic impact with weapon usage and responsible entities.

Insight: The visualization helps identify geographic hotspots and supports location-based risk assessment and targeted response planning.

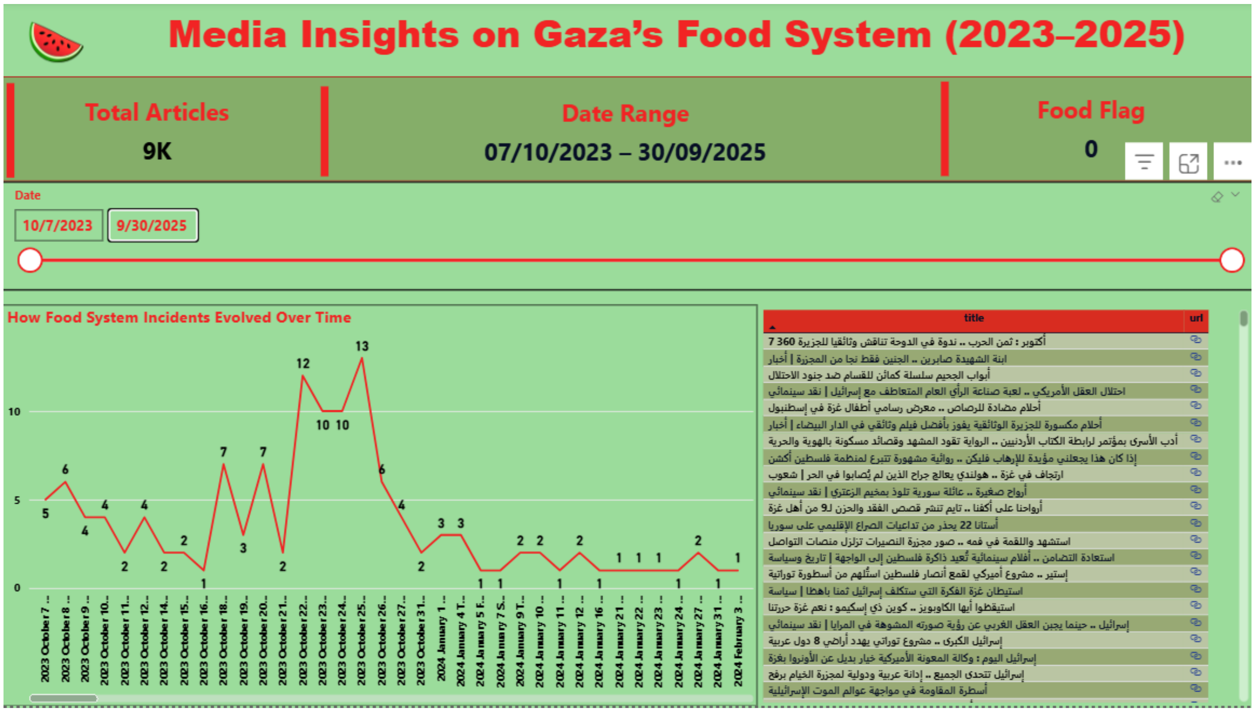

Figure 7: Media Insights on Gaza's Food System (2023-2025)

Data Source and Collection Method

This dashboard is powered by news data retrieved from the **GDELT Project API (Global Database of Events, Language, and Tone)**, an open research platform that indexes and analyzes global news media from trusted sources, including **Al Jazeera**. To build a structured dataset suitable for analysis, a **daily data extraction approach** was implemented covering the selected date range (**07/10/2023 to 30/09/2025**). For each day within this period, the system queries the GDELT **"Doc 2.0"** endpoint and retrieves up to **200 news articles per day** that match a predefined search query, ensuring consistent and systematic coverage of relevant media content.

The query was designed to focus exclusively on Al Jazeera content by filtering the domain (**aljazeera.net**) and targeting Gaza-related and humanitarian keywords in both English and Arabic. This approach ensures that the collected dataset remains relevant to the project scope while minimizing unrelated content.

KPI Cards (Why they matter)

- **Total Articles:** Indicates the total number of collected news articles, reflecting the overall intensity of media coverage.
- **Date Range:** Confirms the active time period selected for analysis.
- **Food Flag:** Displays the number of articles classified as food-system-related, enabling comparison between general Gaza-related news and food-focused coverage.
- **Slicers (Why they matter)**

The dashboard includes a **date range slicer** that allows users to dynamically adjust the timeframe of analysis. This supports:

- Monitoring trends over time (peaks and declines)
- Comparing different periods (before and after key events)
- Focusing analysis on specific weeks or months

Power Query (M) Code Explanation

Figure 8 :Advanced Editor

The custom query was manually written and executed inside Power BI using the Advanced Editor in Power Query. This approach allowed direct control over the data extraction logic and ensured accurate integration of the external news source into the Power BI data model.

```
let
    StartDate = #date(2023, 10, 7),
    EndDate = #date(2025, 9, 30),
    MaxRecordsPerDay = 200,

    SD = Date.From(StartDate),
    ED = Date.From(EndDate),

    QueryText =
        "domain:aljazeera.net (" &
        "gaza OR غزة OR ""قطاع غزة"" OR palestine OR فلسطين OR " &
        """humanitarian situation"" OR ""humanitarian crisis"" OR حصار OR إغاثة OR مساعدات" &
        ")",

    fnFetchDay = (d as date) as table =>
    let
        StartDT = Date.ToText(d, "yyyyMMdd") & "000000",
        EndDT = Date.ToText(d, "yyyyMMdd") & "235959",
```

Figure 9:Advanced Editor1

This section defines the analysis date range (07/10/2023–30/09/2025) and limits daily article retrieval to 200 records for performance control. It also constructs a keyword-based search query restricted to *Al Jazeera*, using relevant Gaza- and humanitarian-related terms in both Arabic and English.

```
Source =
    Json.Document(
        Web.Contents(
            "https://api.gdeltproject.org",
            [
                RelativePath = "api/v2/doc/doc",
                Query = [
                    query = QueryText,
                    mode = "ArtList",
                    format = "json",
                    maxrecords = Text.From(MaxRecordsPerDay),
                    startdatetime = StartDT,
                    enddatetime = EndDT
                ]
            ]
        )
    ),

Articles = try Source[articles] otherwise {},
AsTable = Table.FromList(Articles, Splitter.SplitByNothing(), null, null, ExtraValues.Error),
```

Figure 10:Advanced Editor2

This section sends a request to the GDELT Doc 2.0 API using the predefined query and date range. The response is retrieved in JSON format and converted into a structured object. Error handling is applied to safely manage days with no returned articles, ensuring the data extraction process continues without interruption.

```
        Expanded =
            if Table.RowCount(AsTable) = 0 then
                #table({"Date","seendate","title","url"}, {})
            else
                let
                    E1 = Table.ExpandRecordColumn(
                        AsTable,
                        "Column1",
                        {"seendate","title","url"},
                        {"seendate","title","url"}
                    ),
                    E2 = Table.AddColumn(E1, "Date", each d, type date),
                    E3 = Table.SelectColumns(E2, {"Date","seendate","title","url"})
                in
                    E3
    in
        Expanded,
```

Figure 11:Advanced Editor3

This step checks whether any articles were returned for a given day. If no records exist, an empty table with a consistent schema is created. Otherwise, key fields (`seendate`, `title`, `url`) are expanded from the JSON response, a `Date` column is added to label each article by day, and only the relevant columns are retained to ensure a clean and structured dataset.

```
Days =
    List.Generate(
        () => SD,
        each _ <= ED,
        each Date.AddDays(_, 1),
        each _
    ),

Tables = List.Transform(Days, each fnFetchDay(_)),
Final = Table.Combine(Tables),

ChangedTypes =
    Table.TransformColumnTypes(
        Final,
        {
            {"Date", type date},
            {"seendate", type text},
            {"title", type text},
            {"url", type text}
        }
    ),
```

Figure 12:Advanced Editor4

This section generates a complete list of dates between the start and end of the analysis period. The daily extraction function is applied to each date to retrieve news articles automatically. All daily results are then combined into a single consolidated table, and column data types are standardized to ensure accurate filtering, sorting, and analysis in Power BI.

```
    #"Added Custom" = Table.AddColumn(ChangedTypes, "Custom", each if Text.Contains(Text.Lower([title]), "food")
or Text.Contains(Text.Lower([title]), "غذ")
or Text.Contains(Text.Lower([title]), "قمح")
or Text.Contains(Text.Lower([title]), "طحين")
or Text.Contains(Text.Lower([title]), "خبز")
or Text.Contains(Text.Lower([title]), "مجاعة")
or Text.Contains(Text.Lower([title]), "أمن غذائي")
or Text.Contains(Text.Lower([title]), "food system")
or Text.Contains(Text.Lower([title]), "supply")
then 1 else 0),
    #"Filtered Rows" = Table.SelectRows(#"Added Custom", each true),
    #"Renamed Columns" = Table.RenameColumns(#"Filtered Rows",{{"Custom", "Food_Flag"}})
in
    #"Renamed Columns"
```

Figure 13:Advanced Editor5

This step creates a binary classification column to identify food-system-related news articles. Article titles are scanned for predefined food-related keywords in both Arabic and English. If a match is found, the record is flagged as `1`; otherwise, it is marked as `0`. This classification enables targeted filtering and focused analysis of food-related media coverage.

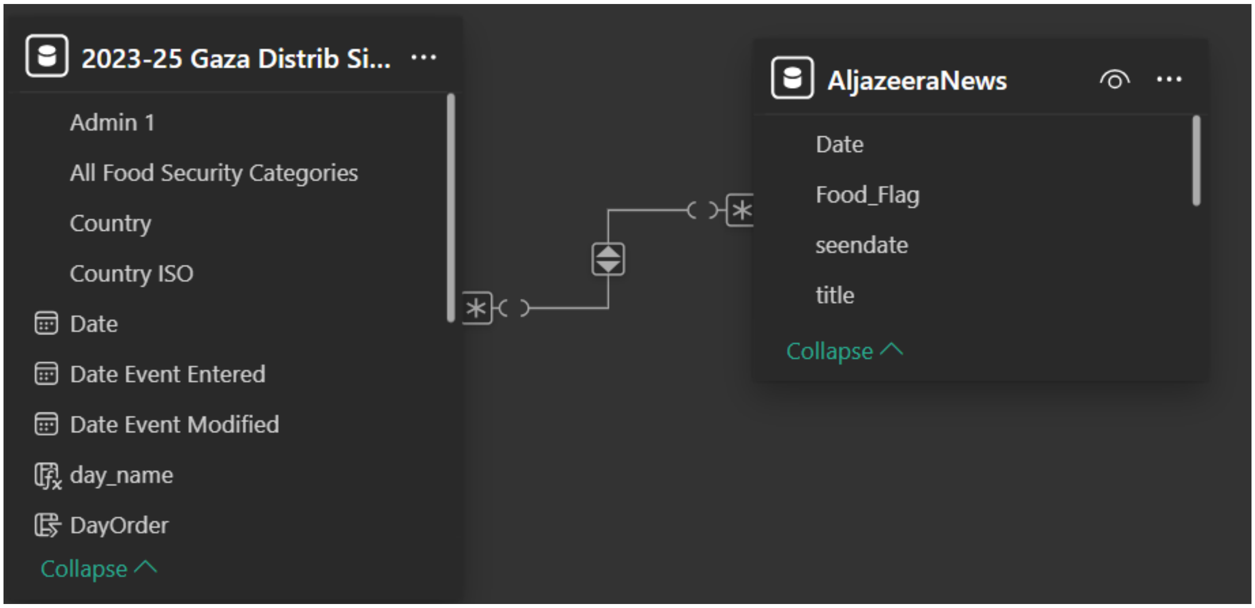

Figure 14:Relationship Design (Many-to-Many)

Relationship Design (Many-to-Many)

A **many-to-many relationship** was implemented between the *Food System* table and the *Al Jazeera News* table to ensure that both datasets remain connected within the dashboard and respond consistently to shared filtering—especially **date-based slicers**. Since the analysis requires the **line chart** and other visuals to reflect the same selected time range, the relationship enables synchronized filtering across both tables. This design ensures that when users adjust the date range, the related media insights and time-series visuals update together, supporting coherent and unified time-based analysis.

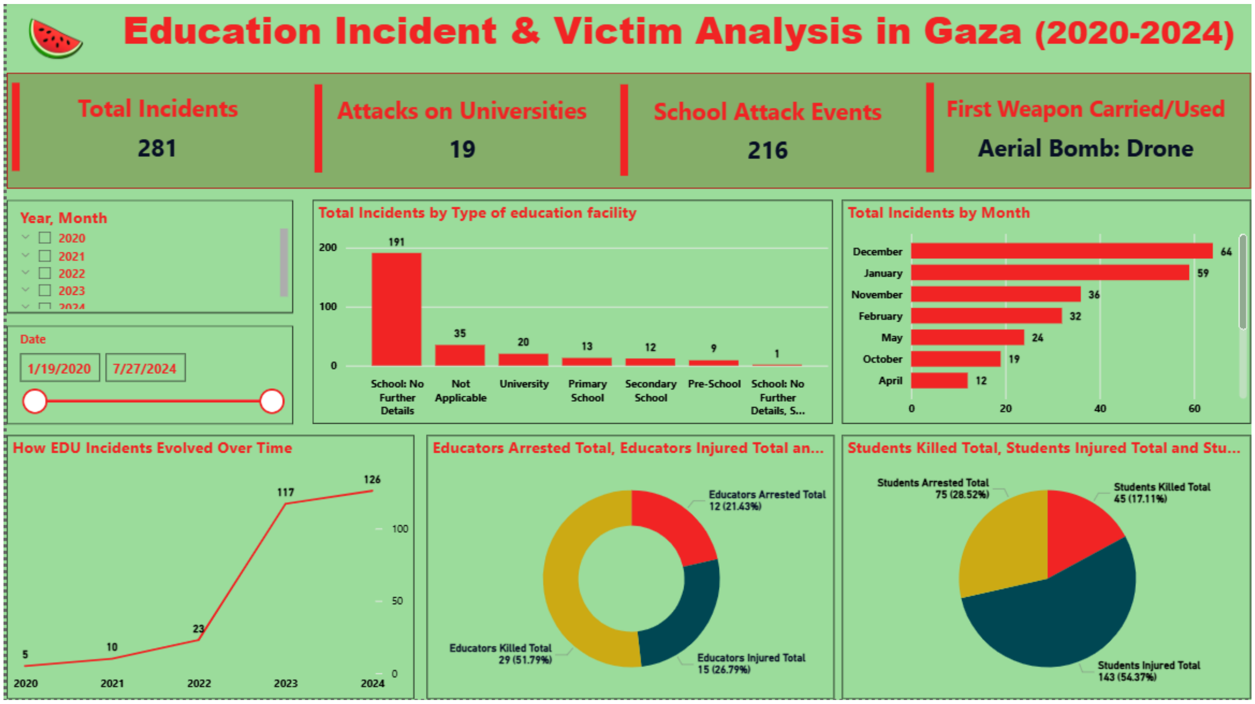

Figure 15:Education Incident & Victim Analysisin Gaza

Dashboard KPI Cards – Education Incidents

The education dashboard includes a set of KPI cards designed to provide a quick and high-level summary of key information related to education-related incidents. These cards enable users to quickly understand the overall situation without navigating detailed charts.

Card 1 – Total Incidents:

Displays the total number of recorded education-related incidents, covering all types of attacks and shelling affecting the education sector.

Card 2 – Attacks on Universities:

Shows the number of incidents that specifically targeted universities, highlighting the level of impact on higher education institutions.

Card 3 – School Attack Events:

Displays the total number of incidents involving attacks on schools, reflecting the scale of direct impact on school-level education.

Card 4 – First Weapon Carried/Used:

Identifies the first weapon type reported in education-related incidents, providing insight into the nature and severity of attacks.

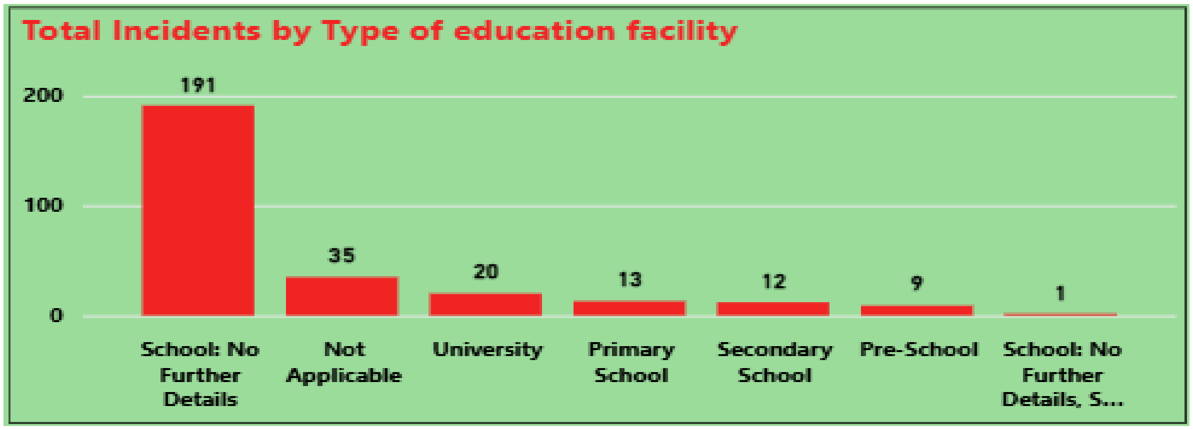

Figure 16:Total Incidents by Type of Education Facility 1

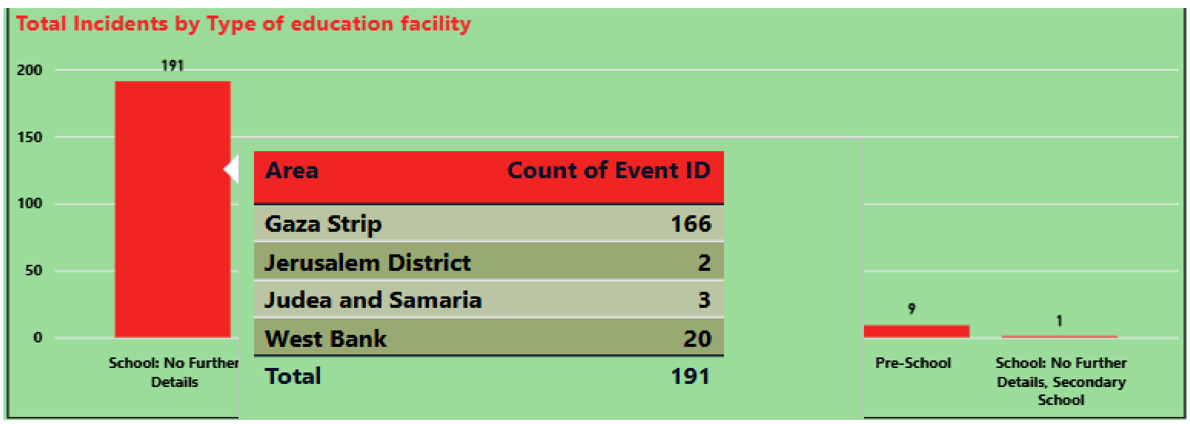

Figure 17:Total Incidents by Type of Education Facility 2

Total Incidents by Type of Education Facility

This bar chart displays the Total Incidents across different types of education facilities, enabling comparison of which facility types were most affected.

An interactive tooltip was added to provide geographic breakdown details. When hovering over each bar, the tooltip shows the distribution by Area (Admin 1 renamed to Area) using Count of Event ID, indicating how many incidents were recorded in each location for the selected facility type

Insight:The results indicate that school-related incidents represent the highest share of total recorded education incidents, highlighting the strong impact of conflict on school-level education.

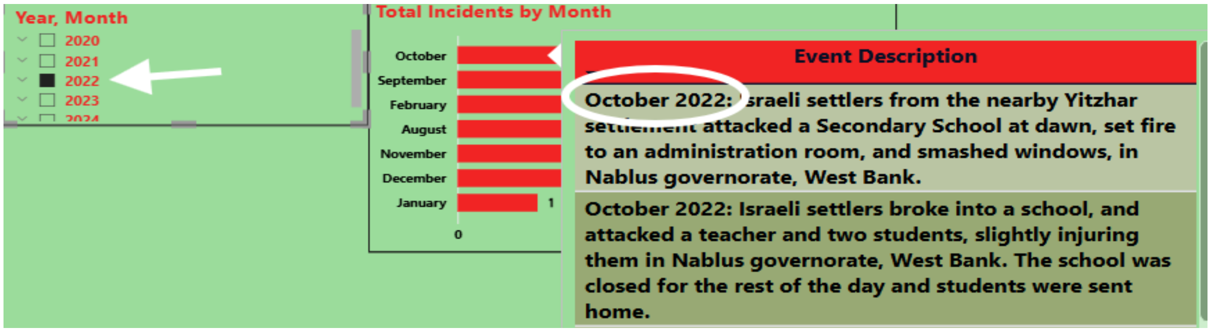

Figure 18:Total Incidents by Month (with Interactive Tooltip)

Total Incidents by Month (with Interactive Tooltip)

This visualization presents Total Incidents by Month, using the Month dimension and Total Incidents as the primary measure. The chart allows users to observe how education-related incidents are distributed across different months within the selected time period.

An interactive tooltip was implemented to provide additional qualitative context. When hovering over a specific month, the tooltip displays the Event Description, which explains the nature of the incident, including details about the attack, location, and affected educational facility. This enhances interpretation by linking quantitative trends with real incident narratives.

The figure also demonstrates the functionality of the Year slicer, where selecting a specific year dynamically updates the monthly incident counts and corresponding event descriptions.

This interaction enables temporal filtering and supports comparative analysis across different years and dates.

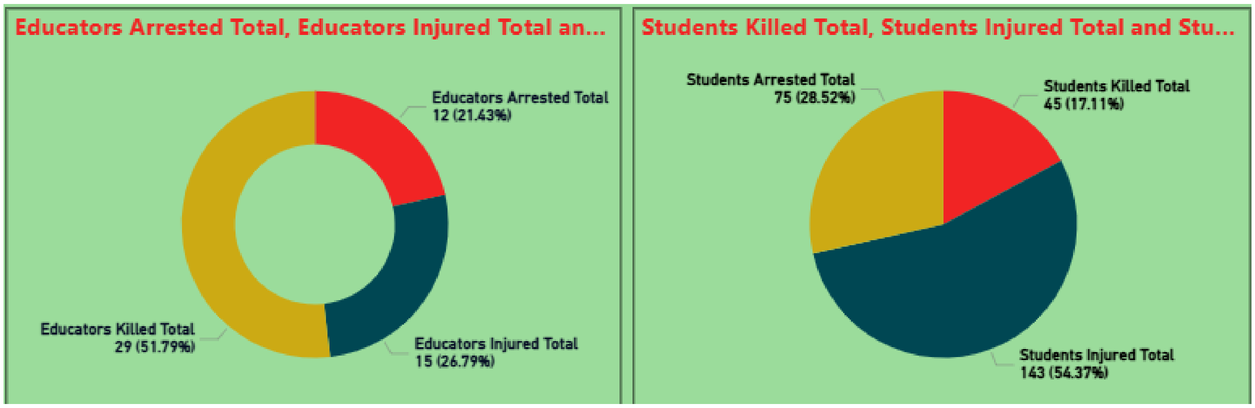

Figure 19:Educators and Students Impact Analysis

Educators and Students Impact Analysis

The **donut chart** and **pie chart** illustrate the distribution of conflict-related impacts on **educators** and **students**, respectively. The visualizations categorize incidents into **killed, injured, and arrested**, allowing a clear comparison of the severity and type of harm experienced by each group.

To accurately represent proportions, **custom measures** were created based on **binary variables (0/1)**, where each value indicates whether a specific impact occurred in an incident. These measures aggregate incidents by category and calculate their percentage share of the total, ensuring consistent and reliable percentage-based representation.

An interactive **tooltip** was also implemented to enhance contextual understanding. When hovering over each segment, the tooltip displays the **location of the event**, providing geographic context and linking human impact outcomes to where the incidents occurred.

Insight:
The charts reveal that **injuries** represent the largest share of recorded impacts for both educators and students, followed by arrests and fatalities. This highlights the significant human cost of conflict on the education sector, affecting both teaching staff and learners.

Advanced Analytics and AI Modeling

This project applies an **NLP-based semantic similarity approach** to identify food security–related news articles from Gaza.
The model combines **semantic similarity analysis** with **keyword-based matching** to capture both implicit and explicit references to food-related topics.
News titles are transformed into numerical representations using a **multilingual sentence embedding model**.
A reference vector representing the food system concept is constructed from predefined food-related terms, and **cosine similarity** is used to measure relevance between each title and the food concept.
An article is classified as food-related if it exceeds a predefined similarity threshold or contains explicit food-related keywords.
The model supports **Arabic and English text** and emphasizes **interpretability**, allowing clear justification of the selection results.

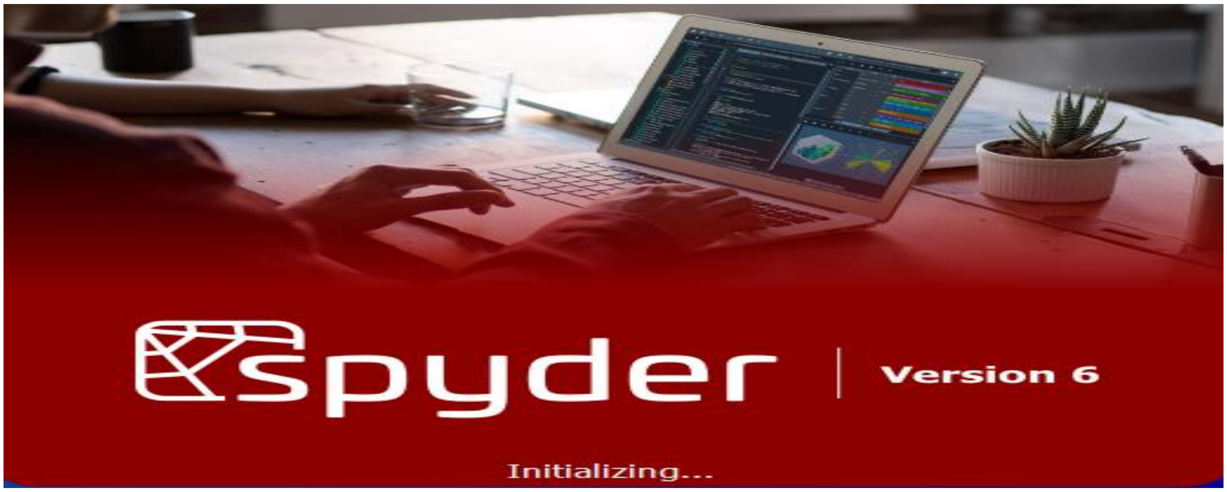

Figure 20:SPYDER

The python code :

```python
import requests
import pandas as pd
from datetime import datetime, timedelta
from sentence_transformers import SentenceTransformer, util
from time import sleep

# CONFIG
START_DATE = datetime(2023, 10, 7)
END_DATE   = datetime(2025, 9, 30)
MAX_RECORDS_PER_DAY = 200
SIMILARITY_THRESHOLD = 0.35
OUTPUT_CSV = "gaza_food_articles_full.csv"
```

Figure 21:python code 1

Imports & Project Configuration

This section initializes the project environment and defines the global configuration parameters.

The imported libraries support data collection, preprocessing, natural language processing, and time handling.

The configuration variables control the time range of the analysis, the maximum number of articles retrieved per day, the semantic similarity threshold, and the output file name.

Defining these parameters at the beginning improves readability, flexibility, and reproducibility of the project.

Usage:

Allows easy adjustment of the analysis period, filtering sensitivity, and output without modifying the core logic.

```python
# GDELT query: only Gaza
QUERY = (
    'domain:aljazeera.net (gaza OR غزة OR "قطاع غزة")'
)

# Food anchors for similarity
FOOD_SYSTEM_ANCHORS = [
"غذاء", "خبز", "قمح", "طحين", "مجاعة", "حصص", "تموين", "مطاعم", "طعام", "مساعدات"
]

# Exact match keywords
EXACT_FOOD_KEYWORDS = [
    "food", "أمن غذائي", "مجاعة", "خبز", "طحين", "قمح", "غذ"
]
```

Figure 22:python code 2

GDELT Query & Keyword Definitions

This section defines the scope of data collection and the criteria used to identify relevant news articles.

The GDELT query restricts the dataset to Al Jazeera articles related to Gaza, ensuring both source and geographic relevance.

Food system anchor terms represent the conceptual meaning of food-related topics, while exact keywords capture articles that explicitly mention food concepts.

This combination supports both semantic understanding and precise keyword matching.

Usage:

Ensures that collected articles are relevant to Gaza and enables accurate detection of food-related content.

```python
# LOAD MODEL
print("Loading multilingual model...")
model = SentenceTransformer("paraphrase-multilingual-MiniLM-L12-v2")
food_embeddings = model.encode(FOOD_SYSTEM_ANCHORS, convert_to_tensor=True)
food_embedding = food_embeddings.mean(dim=0)
```

Figure 23:python code 3

NLP Model Loading & Food Concept Representation

This section loads a multilingual NLP model capable of processing both Arabic and English titles.

Each food anchor term is converted into a numerical embedding, and the mean embedding is computed to create a single “food concept vector.”

This vector acts as the semantic reference point for measuring similarity between the food concept and each news title.

Usage:

Creates a consistent semantic baseline that allows the system to detect food-related news even when exact keywords are not present.

```python
# FETCH FUNCTION
def fetch_day(day):
    start = day.strftime("%Y%m%d") + "000000"
    end   = day.strftime("%Y%m%d") + "235959"

    url = "https://api.gdeltproject.org/api/v2/doc/doc"
    params = {
        "query": QUERY,
        "mode": "ArtList",
        "format": "json",
        "maxrecords": MAX_RECORDS_PER_DAY,
        "startdatetime": start,
        "enddatetime": end,
    }

    try:
        r = requests.get(url, params=params, timeout=30)
        r.raise_for_status()
        data = r.json()
        return data.get("articles", [])
    except Exception as e:
        print(f"Error fetching {day.date()}: {e}")
        return []
```

Figure 24:python code 4

Fetch Function (Daily GDELT API Call)

This function retrieves news articles from the GDELT API for a specific day within the defined time range.

It formats the start and end timestamps in the required GDELT format and sends an API request using the project query and parameters.

A try/except block is used to handle network or API errors gracefully, so the script can continue running without stopping.

Usage:

Provides a reusable daily data extraction method that supports large-scale collection across multiple dates while keeping the system stable during failures.

```python
# COLLECT ARTICLES
records = []
current = START_DATE
print("Collecting Gaza articles from GDELT...")

while current <= END_DATE:
    print(f"Fetching {current.date()}...")
    articles = fetch_day(current)
    for a in articles:
        records.append({
            "date": current.date(),
            "seendate": a.get("seendate"),
            "title": a.get("title"),
            "url": a.get("url"),
        })
    current += timedelta(days=1)
    sleep(0.5)  # small delay to be polite

```

Figure 25:python code 5

Collect Articles Over the Full Date Range

This section runs a day-by-day loop from the start date to the end date to collect Gaza-related articles.

For each day, the script calls fetch_day() and extracts key metadata (date, title, URL, and source).

A small delay is added between requests to reduce API load and follow responsible usage practices.

Usage:

Builds a complete dataset over time and prepares the raw records for cleaning and analysis in the next steps.

```python
# CREATE DATAFRAME & CLEAN
df = pd.DataFrame(records).drop_duplicates(subset=["title", "url"])

# Fix URLs: remove trailing commas/spaces
df["url"] = df["url"].str.rstrip(", \t\n\r")

print(f"Collected {len(df)} Gaza-related articles")

if len(df) == 0:
    print("No articles collected. Exiting.")
    exit()

```

Figure 26:python code 6

Create DataFrame, Remove Duplicates, and Clean URLs

This section converts the collected records into a structured DataFrame for easier processing.

Duplicates are removed using the combination of title and URL to avoid repeating the same article.

URLs are cleaned by removing trailing commas and whitespace to ensure consistency.

A validation step checks if the dataset is empty before continuing to the modeling stage.

Usage:

Improves data quality and prevents incorrect results caused by duplicated or malformed records.

```python
# EXACT FOOD MATCH FLAG
def has_exact_food(title):
    title_lower = title.lower()
    for kw in EXACT_FOOD_KEYWORDS:
        if kw in title_lower:
            return 1
    return 0

df["exact_food_match"] = df["title"].apply(has_exact_food)
```

Figure 27:python code 7

Exact Food Match (Keyword-Based Flag)

This section creates a binary flag that identifies articles with explicit food-related keywords in their titles.

The title is converted to lowercase, and the script checks if any keyword exists within the title text.

If a match is found, the article is labeled as exact_food_match = 1; otherwise, it is 0.

Usage:

Captures clearly food-related articles with high confidence and supports filtering even if semantic similarity scores are low.

```python
# SEMANTIC SIMILARITY
print(f"Encoding {len(df)} titles for similarity...")
title_embeddings = model.encode(df["title"].tolist(), convert_to_tensor=True)

print("Calculating similarity scores...")
scores = util.cos_sim(food_embedding, title_embeddings).squeeze()
df["similarity_score"] = scores.cpu().numpy()
```

Figure 28:python code 8

Semantic Similarity (Embedding-Based Scoring)

This section computes semantic similarity between each news title and the food concept vector.

All titles are converted into embeddings using the same multilingual model.

Cosine similarity is then calculated to produce a numeric score for each article, representing how closely the title relates to the food system concept.

Usage:

Enables detection of food-related articles even when food keywords do not appear directly in the title, improving coverage and relevance.

```python
# FILTER BY SIMILARITY OR EXACT MATCH
filtered = df[(df["similarity_score"] >= SIMILARITY_THRESHOLD) | (df["exact_food_match"] == 1)]
filtered = filtered.sort_values("similarity_score", ascending=False)

# Keep only required columns
filtered = filtered[["date", "seendate", "title", "url", "exact_food_match", "similarity_score"]]
```

Figure 29:python code 9

Filtering Logic (Threshold OR Exact Match)

This section selects the final set of food-related articles using two criteria:

Titles with a semantic similarity score above the defined threshold, OR

Titles that contain explicit food keywords (exact match).

The results are then sorted by similarity score to prioritize the most relevant articles, and only the required columns are kept for reporting.

Usage:

Creates a reliable final dataset by combining semantic detection with keyword confirmation and produces an organized output ready for analysis or visualization.

```python
# EXPORT
filtered.to_csv(OUTPUT_CSV, index=False, encoding="utf-8-sig")
print(f"Saved {len(filtered)} records to {OUTPUT_CSV}")
print("Done.")
```

Figure 30:python code 10

Export Results to CSV

This section exports the filtered dataset to a CSV file using UTF-8 encoding to preserve Arabic characters.

The exported file contains the final list of food-related Gaza articles, including the similarity score and the exact match flag for transparency and interpretability.

Usage:

Produces a shareable output that can be used in Power BI dashboards, reporting, or further machine learning analysis.

The following table summarizes the Python packages used in this project and their respective purposes.

Package	Purpose
requests	API requests to GDELT
pandas	Data processing & CSV output
sentence-transformers	NLP semantic embeddings
torch	Deep learning backend
datetime	Date handling (built-in)
time	Execution delays (built-in)

Figure 31: :python packages

The table below shows the final output of the hybrid information retrieval model.

date	seendate	title	url	exact_food_match	similarity_score
8/11/2025	20250811T131500Z	أنس الذي وفى	https://www.aljazeera.net/blogs/2025/8/11/%D8%A3%D9%86%D8%B3-%D8%A7%D9%8	0	0.67418516
4/6/2024	20240406T031500Z	طبّاخ السمّ يذوقه في مطبخ الإبادة العالمي !	https://www.aljazeera.net/opinions/2024/4/6/%D8%B7%D8%A8%D8%A7%D8%AE-%D8	0	0.6270249
5/8/2025	20250508T084500Z	في رثاء سايكس بيكو والتعايش الهش	https://www.aljazeera.net/blogs/2025/5/7/%D9%81%D9%8A-%D8%B1%D8%AB%D8%A7	0	0.61915904
6/10/2024	20240610T160000Z	فضائل عابرة !	https://www.aljazeera.net/blogs/2024/6/10/%D9%81%D8%B6%D8%A7%D8%A6%D9%84	0	0.6057997
4/8/2024	20240408T221500Z	في رثاء عَمرو	https://www.aljazeera.net/opinions/2024/4/9/%D9%81%D9%8A-%D8%B1%D8%AB%D8	0	0.6015823
2/23/2024	20240223T230000Z	تعرف على أبرز المواثيق الدولية المقرّة للحق في الغذاء	https://www.aljazeera.net/programs/2024/2/24/%D8%AA%D8%B9%D8%B1%D9%81-%D	1	0.58922315
6/4/2025	20250604T104500Z	طفل غزي لتايمز : الطعام الذي كانت أمي ستحضره لنا تضرج بدمائها	https://www.aljazeera.net/politics/2025/6/4/%D8%B7%D9%81%D9%84-%D8%BA%D8%	0	0.5863863
5/28/2025	20250528T221500Z	برنامج الأغذية العالمي : جحافل من الجوعى اقتحمت أحد مستودعاتنا بغزة	https://www.aljazeera.net/news/2025/5/29/%D9%82%D8%AA%D9%8A%D9%84%D8%A	1	0.5687518
6/30/2024	20240630T203000Z	عن الطوفان ومجرى القَصية !	https://www.aljazeera.net/blogs/2024/6/30/%D8%B9%D9%86-%D8%A7%D9%84%D8%B	0	0.5603734
8/31/2025	20250831T213000Z	جدعون ليفي : مأساة أن نغضب لأسير جائع لكن نشمت بمقتل أطفال ينتظرون الخبز	https://www.aljazeera.net/politics/2025/8/31/%D8%AC%D8%AF%D8%B9%D9%88%D9%	1	0.5593687
10/14/2024	20241014T084500Z	قصة العشاء الأخير لجنود لواء غولاني الإسرائيلي	https://www.aljazeera.net/news/2024/10/14/%D9%82%D8%B5%D8%A9-%D8%A7%D9%	0	0.5591217
6/1/2025	20250601T110000Z	مجزرة رفح .. خبز مغمس بالدم ومساعدات تتحول لأفخاخ موت	https://www.aljazeera.net/news/2025/6/1/%D9%85%D8%AC%D8%B2%D8%B1%D8%A9-	1	0.5576142
5/15/2025	20250515T154500Z	الجوع والخوف : أدوات استعمارية موجهة	https://www.aljazeera.net/blogs/2025/5/15/%D8%A7%D9%84%D8%AC%D9%88%D8%B	0	0.5467762
7/29/2025	20250729T014500Z	الجوع على مائدة النقاش بقمة الأمم المتحدة للغذاء بأديس أبابا	https://www.aljazeera.net/politics/2025/7/29/%D8%A7%D9%84%D8%AC%D9%88%D8%	1	0.5441686
10/16/2024	20241016T143000Z	براغماتيون ولكن ..	https://www.aljazeera.net/blogs/2024/10/16/%D8%A8%D8%B1%D8%A7%D8%BA%D9%8	0	0.5338585
7/22/2025	20250722T184500Z	حين يصبح الخبز سلاحًا : التجويع كإستراتيجية لإخضاع غزة	https://www.aljazeera.net/blogs/2025/7/22/%D8%AD%D9%8A%D9%86-%D9%8A%D8%B	1	0.5301553
7/20/2025	20250720T221500Z	الأغذية العالمي : شخص من كل 3 بغزة لا يتناول الطعام لمدة أيام	https://www.aljazeera.net/news/2025/7/21/%D8%B9%D8%A7%D8%AC%D9%84-%D8%A	1	0.52986574
8/8/2024	20240808T034500Z	المطبخ العالمي : مقتل أحد موظفينا في غزة	https://www.aljazeera.net/news/2024/8/8/%D8%A7%D9%84%D9%85%D8%B7%D8%A8	0	0.52630544
3/13/2024	20240313T071500Z	الغذاء لغزة .. إليك تفاصيل المبادرة الإيطالية الأمنية المشتركة	https://www.aljazeera.net/politics/2024/3/13/%D8%A7%D9%84%D8%BA%D8%B0%D8%	1	0.52497053
4/26/2024	20240426T193000Z	أطباق فلسطينية غيرتها الحرب وأمهات يبدعن في توفير الطعام	https://www.aljazeera.net/lifestyle/2024/4/26/%D8%A3%D8%B7%D8%A8%D8%A7%D9%	0	0.5247383
3/5/2024	20240305T150000Z	رحلة البحث عن طعام داخل أسواق مدينة غزة | سياسة	https://www.aljazeera.net/politics/2024/3/5/%D8%B1%D8%AD%D9%84%D8%A9-%D8%	0	0.52094
8/25/2024	20240825T124500Z	الجزء المجهول من تأثير السنوار !	https://www.aljazeera.net/blogs/2024/8/25/%D8%A7%D9%84%D8%AC%D8%B2%D8%A	0	0.51963466
6/3/2025	20250603T100000Z	الطعام فخ للقتل .. شهادات مروعة عن مجازر إسرائيل بغزة	https://www.aljazeera.net/news/2025/6/3/%D8%A7%D9%84%D8%B7%D8%B9%D8%A7	0	0.5194578
7/25/2025	20250725T133000Z	غزاوي بلندن : أعيش جحيما فثلاجتي ممتلئة وأمي مُجوّعة وأختي تطعم أطفالها الحشائش	https://www.aljazeera.net/politics/2025/7/25/%D8%BA%D8%B2%D8%A7%D9%88%D9%	0	0.5170629

gaza_food_articles_full

Figure 32: table data set

Introduction to the Results Dataset

The attached dataset represents the **final output** of an NLP-based analytical model designed to identify **food security–related news articles from Gaza**.

The dataset includes article titles, URLs, observation dates, and two key indicators that explain **why each article was selected** as relevant to the food system topic.

The model integrates **semantic similarity analysis** with **keyword-based matching**, allowing it to capture both explicit and implicit references to food-related issues.

This hybrid approach improves coverage while maintaining interpretability and transparency in the selection process.

Comparison Between exact_food_match and similarity_score

1) exact_food_match

exact_food_match is a binary indicator (0/1) that identifies articles whose titles contain explicit food-related keywords, such as bread, wheat, flour, famine, or food security.

- A value of **1** indicates the presence of at least one predefined food keyword.
- A value of **0** indicates no direct keyword match.

Strengths:

- Simple and easy to interpret.
- High precision when food-related terms are explicitly mentioned.

Limitations:

- May miss food-related articles that do not contain explicit keywords.
- May include articles that mention keywords in a non-food context (e.g., medical or humanitarian aid).

2) similarity_score

similarity_score is a **continuous numerical value** that represents the **semantic closeness** between a news title and the food system concept.
It is computed using **sentence embeddings** and **cosine similarity**, allowing the model to evaluate meaning rather than exact wording.

- Higher values indicate stronger semantic relevance to food-related topics.
- Articles exceeding a predefined threshold are classified as food-related.

Strengths:

- Captures implicit food-related references.
- Works across Arabic and English content.

Limitations:

- Requires threshold tuning.
- May assign moderate similarity to contextually related but non-food articles.

Key Difference

- **exact_food_match answers:** “Does the title explicitly mention a food-related term?”
- **similarity_score answers:** “Is the meaning of the title related to food security?”

Using both indicators together allows the system to balance **precision and recall**.

	1	0
1	True Positives (TP)	False Positives (FP)
0	False Negatives (FN)	True Negatives (TN)

Figure 33:Classification Logic and Evaluation (TP / TN / FP / FN)

Classification Logic and Evaluation (TP / TN / FP / FN)

The final classification task determines whether an article is **food-related (1)** or **not food-related (0)**.
Evaluation is based on comparing the model’s output with a **manual ground truth assessment**.

Definitions:

- **True Positive (TP):** Article classified as food-related and confirmed as food-related.
- **True Negative (TN):** Article classified as non-food-related and confirmed as non-food-related.
- **False Positive (FP):** Article classified as food-related but not actually related to food.
- **False Negative (FN):** Article classified as non-food-related but actually related to food.

Accuracy Consideration

As this model is built on a **hybrid information retrieval framework** rather than supervised machine learning, conventional accuracy metrics based on labeled training datasets are not directly applicable.
Instead, the model's performance is evaluated through **manual validation of representative samples**, where classification outcomes are analyzed using true positives, true negatives, false positives, and false negatives.

Accuracy-related behavior is primarily governed by the **semantic similarity threshold**, which determines the sensitivity of the model to food-related concepts, as well as the **interaction between keyword-based precision and semantic recall**.
Adjusting this threshold allows fine-tuning of the trade-off between capturing a broader set of relevant articles and minimizing irrelevant selections.

Furthermore, model accuracy can be significantly enhanced by **expanding the textual scope of analysis**.
Currently, the model relies on news titles, which may provide limited semantic context. Incorporating **article descriptions, abstracts, or full-text content** would enrich contextual understanding, reduce ambiguity, and improve the reliability of similarity scoring.

In addition, iterative refinement of the **keyword list** and periodic review of misclassified cases (false positives and false negatives) can further strengthen model robustness.
If labeled data becomes available in future iterations, the hybrid retrieval approach can be complemented with **supervised classification techniques**, enabling quantitative accuracy measurement while preserving interpretability.

Overall, this methodology offers a **transparent, flexible, and scalable evaluation framework**, allowing continuous accuracy improvement even in the absence of labeled training data.

Tools Research and Selection Effort

Several tools were explored and selected to support different stages of the project workflow.

Microsoft Excel was used in the initial stage for data inspection, cleaning, and manual validation of the extracted results, allowing quick review and verification of the dataset.

For data visualization and presentation, **Power BI** was utilized to create interactive dashboards and visual summaries that help analyze trends and patterns in food security–related news. This supported clearer interpretation and communication of the results.

Python was chosen as the core implementation language due to its strong capabilities in data analysis and natural language processing. The code was developed and executed using the **Spyder IDE**, which provided an efficient environment for writing, testing, and debugging the model.

In addition, **artificial intelligence tools** were used to support research, code organization, and documentation, improving productivity and enhancing the overall quality of the project.

Project Deployment Effort – Use Case

In this project, the data was obtained from the **Humanitarian Data Exchange (HDX)** platform through the website
https://data.humdata.org, which is a reliable source of open humanitarian data.
The relevant datasets were extracted and downloaded in **Excel format** to support subsequent processing and analysis stages.
Following data extraction, **Power BI** was used to visualize and analyze the data in a clear and intuitive manner, enabling effective communication with **business users and decision-makers** interested in monitoring food security–related issues.
Two separate Power BI files were developed:

- A file focused on **Food (Food Security)**
- A file focused on **Education**

This separation allowed for more structured analysis and domain-specific dashboards, providing interactive visualizations that help users explore trends and patterns within each topic independently.

After completing the visualization and exploratory analysis phase, the project proceeded to the **code implementation stage** using **Python**.
A Natural Language Processing (NLP)–based model was developed to process and classify news articles related to food security.
The code performs text processing, computes semantic similarity scores, and generates structured output files that can be reused for further analysis or visualization.

This integrated deployment approach enables users to **efficiently review large volumes of data and news articles**, reduces reliance on manual analysis, and supports informed decision-making through clear and interpretable model outputs.

- **Sequential Implementation Steps**

1. Defined the project objective and scope, focusing on food security and education–related analysis.
2. Collected datasets from the **Humanitarian Data Exchange (HDX)** platform.
3. Extracted and stored the data in **Excel files**.
4. Imported the data into **Power BI** and created two separate analytical files (Food and Education).
5. Designed interactive dashboards to visualize trends and patterns.

6. Developed Python code to process textual data and implement the NLP-based model.
7. Applied semantic similarity and keyword-based classification techniques.
8. Generated structured output files to support analysis, visualization, and decision-making.

Results

Based on the Power BI dashboards built from the HDX incident datasets (food systems incidents in Gaza and education-related incidents), the analysis reveals clear time-based escalation patterns and location-based clustering of impact. In the food systems dataset, incident levels fluctuate significantly over the analyzed period, with distinct peaks indicating phases of intensified disruption—particularly affecting food aid systems, which emerge as the most frequently impacted component compared to other food system elements. In the education dataset, the distribution by facility type shows that school-related incidents account for the highest proportion, highlighting how conflict pressure is primarily concentrated on school-level education and results in ongoing disruption for students and educators.

The two most important visuals used in this project are the line chart (incidents over time) and the map (geographical distribution). The line chart is especially effective for decision-making, as it clearly illustrates when escalation occurs, highlights peaks and declines over time, and enables comparison across different periods using consistent slicers. In addition, the line chart allows direct access to the news related to a specific day by selecting a peak in incidents, which supports deeper contextual analysis of the events reported on that date and strengthens the linkage between quantitative trends and qualitative news content. The map is equally critical, as it shows where incidents are geographically concentrated, transforming incident records into actionable spatial hotspots. When combined with filters such as weapon type or perpetrator, the map supports targeted interpretation and location-based planning, including prioritizing areas for monitoring, support, and intervention. Together, these two visuals provide the most comprehensive insight, as they address the two core questions in humanitarian analysis: **when did the escalation intensify, and where was its impact most severe?**

References

1. **Humanitarian Data Exchange (HDX).**
   Incident datasets related to food systems and education in Gaza.
   Retrieved from: https://data.humdata.org
2. **Al Jazeera.**
   News articles used for contextual analysis and daily incident interpretation.
   https://www.aljazeera.net